

Coherent Sources and Young's Double Slit

B.Sc. (V Semester) — Physical Optics · Kumaon University
Unit II: Interference · Lecture 2 · Instructor **Dr. Prateek Sharma**

Coherent sources · Path and phase difference · Fringe width · Conditions for good fringes · Analytical treatment

1 Coherent Sources

Two sources of light are said to be **coherent** if they emit continuous light waves of the *same frequency (or wavelength)* and *same amplitude*, with either the same phase or a *constant phase difference*.

Production of coherent sources

In practice, two coherent sources are always derived from a *single* source, so that any random phase change is common to both. The usual methods are:

- (i) A real narrow source and its virtual image produced by reflection — as in *Lloyd's single mirror*.
- (ii) Two virtual images of the same source produced by reflection — as in *Fresnel's double mirror*.
- (iii) Two virtual images of the same source produced by refraction — as in *Fresnel's biprism*.
- (iv) Two real images of the same source produced by refraction — as in *Billet's split lens*.
- (v) By dividing the amplitude of a portion of the wavefront by reflection, refraction or both — as in *Newton's rings*, the *Michelson interferometer* and *thin films*.

2 Path Difference, Phase Difference and Fringe Width

The previous section obtained the conditions for maxima and minima in terms of the *path difference*, while the intensity was expressed in terms of the *phase difference* ϕ . The two are directly related.

2.1 Relation between phase difference and path difference

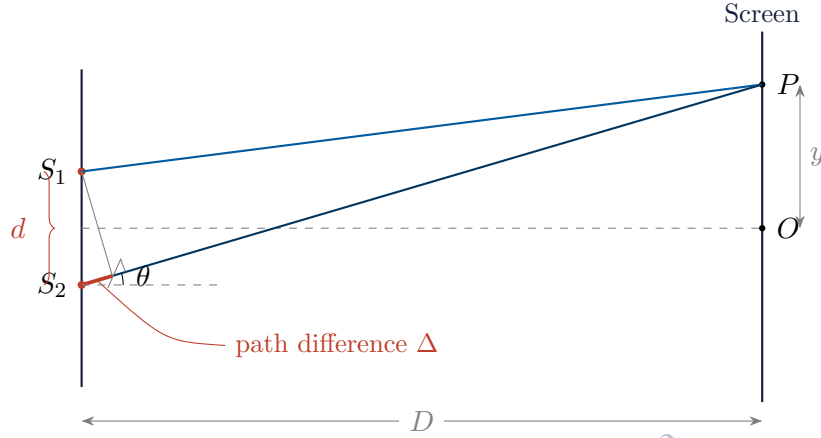
A path difference of one full wavelength λ corresponds to a phase difference of 2π . Hence a path difference Δ produces a phase difference

$$\phi = \frac{2\pi}{\lambda} \Delta$$

This bridges the two descriptions: the bright condition $\phi = 2n\pi$ gives $\Delta = n\lambda$, and the dark condition $\phi = (2n+1)\pi$ gives $\Delta = (2n+1)\frac{\lambda}{2}$, this result will be proved in subsequent sections.

2.2 Geometry of Young's double slit

Let two coherent slits S_1 and S_2 , separated by a small distance d , be illuminated by a monochromatic source. The interference pattern is observed on a screen placed at a distance D ($D \gg d$). Let O be the point on the screen directly opposite the mid-point of the slits, and P a point at distance y from O .



2.3 Path difference

The path difference between the two waves reaching P is $\Delta = S_2P - S_1P$. From the geometry,

$$S_1P^2 = D^2 + \left(y - \frac{d}{2}\right)^2, \quad S_2P^2 = D^2 + \left(y + \frac{d}{2}\right)^2.$$

Subtracting,

$$S_2P^2 - S_1P^2 = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2 = 2yd,$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2yd.$$

Since $D \gg d, y$, both paths are nearly D , so $S_2P + S_1P \approx 2D$. Therefore

$$\Delta = S_2P - S_1P = \frac{yd}{D}$$

Equivalently, from the small triangle in the figure, $\Delta = d \sin \theta \approx d \tan \theta = \frac{yd}{D}$.

2.4 Positions of bright and dark fringes

Bright fringes occur where $\Delta = n\lambda$:

$$\frac{y_n d}{D} = n\lambda \Rightarrow \boxed{y_n = \frac{n\lambda D}{d}}, \quad n = 0, 1, 2, \dots$$

Dark fringes occur where $\Delta = (2n + 1)\frac{\lambda}{2}$:

$$\frac{y_n d}{D} = (2n + 1)\frac{\lambda}{2} \Rightarrow \boxed{y_n = (2n + 1)\frac{\lambda D}{2d}}, \quad n = 0, 1, 2, \dots$$

2.5 Fringe width

The fringe width β is the separation between two consecutive bright (or two consecutive dark) fringes:

$$\beta = y_{n+1} - y_n = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d},$$

$$\beta = \frac{\lambda D}{d}$$

This confirms the fringe-width formula quoted earlier. Note that all fringes are equally spaced, and the spacing is the same whether measured between bright or between dark fringes.

3 Conditions for a Good Interference Pattern

- (a) Observability
- (b) Good contrast of fringes
- (c) Stationary interference pattern

3.1 Observability

Let d be the separation of the two coherent sources, λ the wavelength, and D the distance between the sources and the screen. The fringe width is

$$\beta = \frac{D\lambda}{d}$$

For the fringes to be resolvable and observable:

- (i) d should be small,
- (ii) D should be large,
- (iii) the background should be dark, for good visibility.

3.2 Good contrast

The amplitudes of the interfering waves should be nearly equal, since

$$I_{\max} \propto (a_1 + a_2)^2, \quad I_{\min} \propto (a_1 - a_2)^2.$$

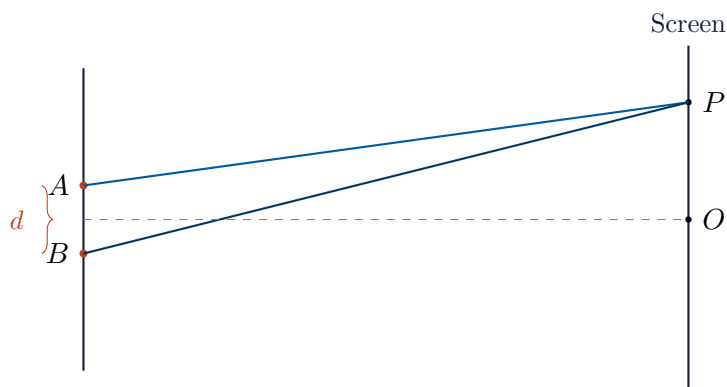
If $a_1 = a_2$, then $I_{\min} = 0$ and the dark fringes are truly dark, giving the sharpest contrast.

3.3 Stationary interference pattern

- (i) The sources must be coherent, i.e. emit continuous waves of the same wavelength with the same phase or a constant phase difference.
- (ii) The sources must emit waves of the same wavelength and period.

4 Analytical Treatment of Interference

Consider two coherent sources A and B sending waves to a point P on the screen. The wave from B travels an extra path x compared with the wave from A .



Taking the displacement of the wave from A at P as

$$y_1 = a \sin \frac{2\pi}{\lambda} vt,$$

and that from B , delayed by the extra path x , as

$$y_2 = a \sin \frac{2\pi}{\lambda} (vt + x),$$

where λ is the wavelength and v the wave speed, the resultant displacement is

$$y = y_1 + y_2 = a \sin \frac{2\pi}{\lambda} vt + a \sin \frac{2\pi}{\lambda} (vt + x).$$

$$\text{Using } \sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2},$$

$$y = 2a \cos \left(\frac{2\pi}{\lambda} \cdot \frac{x}{2} \right) \sin \frac{2\pi}{\lambda} \left(vt + \frac{x}{2} \right).$$

The resultant amplitude is therefore

$$A = 2a \cos \frac{\pi x}{\lambda}.$$

Minima (dark fringes)

The amplitude vanishes when

$$\cos \frac{\pi x}{\lambda} = 0 \Rightarrow \frac{\pi x}{\lambda} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots, (2n+1)\frac{\pi}{2},$$

$$x = (2n+1)\frac{\lambda}{2}$$

The intensity is minimum when the path difference is an *odd* multiple of half the wavelength.

Maxima (bright fringes)

The amplitude is greatest when

$$\cos \frac{\pi x}{\lambda} = 1 \Rightarrow \frac{\pi x}{\lambda} = 0, \pi, 2\pi, \dots, n\pi,$$

$$x = n\lambda$$

The intensity is maximum when the path difference between the two wave trains is an integral multiple of the wavelength (equivalently, an even multiple of $\lambda/2$).